

Navigating Context Collapse: How Network Structure and Message Directedness Shape Online Expression

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Abstract

Social media platforms present users with persistent, publicly visible audiences composed of multiple social groups. This convergence of social contexts—referred to as context collapse—limits users’ ability to tailor communication to specific audiences. While prior research has emphasized users’ perceptions and strategies, there is limited behavioral evidence linking audience structure to observable expression patterns. Using a panel dataset of 1,827 U.S.-based Twitter users, we investigate how network structure and message directedness relate to variation in tweet content over time. Our findings reveal that directed tweets consistently contain less offensive and sensitive content, more positive sentiment, and less negative sentiment than undirected tweets. Network size shows suppression effects within users: as individuals’ networks grow, they post less offensive, sensitive, and negative content while increasing positive expression. Meanwhile, dense networks are associated with more offensive content and negative sentiment. Longitudinal analysis demonstrates systematic behavioral adaptation, with users exhibiting declining offensive, sensitive, and negative content over time. Our results offer implications for platform design, highlighting how structural and interactional cues could help users navigate audience complexity.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing**.

Keywords

Context Collapse; Self-Presentation; Twitter; Social Networks; Audience Design

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1 Introduction

Social media platforms present users with persistent, publicly visible audiences composed of multiple social groups. The resulting convergence of social contexts—referred to as context collapse—limits users’ ability to tailor communication to specific audiences. To manage this complexity, users may adjust how they frame their communication and whom they address. Prior research on context collapse has emphasized the role of imagined audiences [10, 11], privacy strategies [15, 19], and self-censorship [7, 14]. While these studies offer insight into users’ perceptions and strategies, they rely largely on qualitative methods and are limited in empirical evidence linking audience structure to observable outcomes.

Self-presentation theory conceptualizes individuals as performers who tailor their self-image to audience expectations [8]. In face-to-face settings, audiences are typically well-bounded and contextually constrained, making strategic impression management easier. Online platforms, by contrast, blur social boundaries and complicate audience segregation, often forcing users to speak simultaneously to friends, coworkers, acquaintances, and strangers [1, 11, 15, 19]. These collapsed contexts increase the risk of unintended exposure, misinterpretation, or backlash, particularly when content intended for one audience becomes visible to another [18]. To manage these risks, users often employ adaptive strategies grounded in their imagined audience—a mental model of who is watching [10, 11]. These strategies include posting for the “lowest common denominator” to minimize offense across audience segments [9, 14], avoiding sensitive topics altogether [7], or tailoring content to match shared interests or norms [17].

In this study, we focus on two key factors of audience configuration: a user’s network structure (size and density) and the “directedness” of their messages. We refer to tweets that include @mentions or replies as *directed tweets* and those without as *undirected tweets*. Drawing from theories of network influence and audience design, we examine how these features shape tweet behavior across multiple dimensions.

2 Theoretical Framework

Large social networks typically expose users to more heterogeneous audiences and increase the potential reach of communication. Prior work suggests this can create reputational risk and inhibit norm violations, especially in public forums [17]. Dense networks—where audience members are themselves more connected—may amplify pressures of impression management, as overlapping ties increase the likelihood that norm violations will be noticed and judged by multiple close contacts [4]. Alternatively, larger audiences may encourage more expressive or emotionally charged content if users seek attention or engagement.

Tweets that directly mention or reply to others are formally public but functionally interpersonal. This distinction between directed and undirected tweets builds on audience design theory [3, 6], which posits that communication is shaped by the imagined audience. When users direct tweets at specific others, they are more likely to imagine a specific, concrete audience. This heightened audience salience may prompt more moderated tweets. In contrast, undirected tweets resemble generalized broadcasts to an indeterminate public audience, which may diminish perceived social stakes and enable less moderated expression.

The temporal dimension adds further complexity to these dynamics. Users may develop more sophisticated audience management strategies through platform experience, potentially leading to systematic changes in expression patterns over time. Understanding how network properties and communication patterns jointly influence expression has implications for both platform design and broader questions about online discourse quality.

This study investigates the following research questions:

RQ1: How does message directedness influence tweet content? We examine whether directed tweets differ systematically from undirected tweets in terms of offensive content, sensitive content, and emotional expression across positive, negative, and neutral sentiment.

RQ2: How do network structural properties relate to variation in tweet content? We investigate the relationships between network size and tweet content patterns, and between network density and tweet content patterns.

RQ3: How do users adapt their tweet content over time? We examine longitudinal trends in offensive content, sensitive content, and sentiment to understand behavioral adaptation to audience structure and platform norms.

3 Methodology

This study investigates how social network structure shapes online communication behavior using a longitudinal panel of Twitter activity. By lagging network features relative to tweet content, we examine how changes in network structure—specifically, network size and density—influence the volume, targeting, and content characteristics of tweets over time.

The dataset is derived from a random sample of 20,000 users drawn from a larger pool of 26 million U.S.-based Twitter users originally collected via a snowball sample between 2013 and 2014 [13]. To ensure our analyses reflect meaningful and stable patterns of social interaction, we filtered for users with at least 100 followers, more than 200 posted tweets, and at least three years of observable activity. This yielded a final sample of 1,827 ego users, for whom we obtained up to 3,200 of their most recent tweets.

Dynamic ego networks were constructed based on bidirected @mentions, with an undirected edge added when two users mutually mentioned each other. From these evolving networks, we computed two structural metrics: *network size* (number of alters) and *network density* (proportion of edges among alters). Tweet content was analyzed using BERT-based classifiers from TweetNLP [5]. We labeled each tweet with directedness (based on the presence of @mentions or replies) and three content dimensions—offensiveness, sensitivity, and sentiment. These features enabled construction of

a panel dataset, capturing both structural and tweet attributes over time.

Our regression analysis proceeded in several stages to systematically examine different aspects of tweet behavior. We first modeled tweet activity patterns, examining how network structure influences tweet volume and the proportion of directed tweets. We then analyzed tweet content using two complementary model types: (1) within-user models that employed fixed effects estimation to analyze how individual users' tweet behavior changed over time in relation to their own network evolution, controlling for time-invariant characteristics, and (2) between-user models that examined how average differences in network structure across users related to variation in their tweet behavior. In these models, we also examined temporal trends to assess behavioral adaptation over time. This approach enabled us to distinguish within-user behavioral adaptation from broader differences between users.

Independent variables included network size, network density, tweet directedness, and time. For each user-month, network variables were calculated based on the previous month's ego network, then used to predict tweet outcomes in the subsequent month. The purpose of this one-month lag is to reduce reverse causality concerns. Dependent variables captured tweet volume, proportion of directed tweets, and proportions of offensive, sensitive, and sentiment-labeled tweets.

4 Results

We present findings from panel regressions examining how network structure and message directedness relate to user tweet behavior over time.

Table 1: Network Structure Effects on Tweet Activity

Variable	Tweet Volume		Directed Proportion	
	Between	Fixed	Between	Fixed
Network Size	0.361***	0.072***	0.345***	0.234***
Network Density	-0.076**	0.069***	0.002	0.044***
Constant	0.366***	–	0.200***	–
R^2	0.294	0.007	0.189	0.044
F-statistic	380.5	240.2	212.8	1,671

Significance: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

As summarized in Table 1, we first analyzed network effects on tweet activity patterns. Network size demonstrated consistent positive associations with both tweet volume and directedness across all models. Users with larger networks posted more frequently ($\beta = 0.361$, $p < 0.001$ between-user; $\beta = 0.072$, $p < 0.001$ within-user) and directed a higher proportion of their tweets to specific users ($\beta = 0.345$, $p < 0.001$ between-user; $\beta = 0.234$, $p < 0.001$ within-user). Meanwhile, network density revealed contrasting patterns. Between users, higher density was associated with reduced tweet volume ($\beta = -0.076$, $p < 0.01$), while within-user analysis showed density increases predicted higher tweet volume ($\beta = 0.069$, $p < 0.001$) and greater directedness ($\beta = 0.044$, $p < 0.001$).

We then further examined how network structure and message directedness influence tweet content, the results of which are reported in Tables 2 and 3.

Message directedness was consistently associated with more restrained communication across all content dimensions. Directed tweets contained significantly less offensive content ($\beta = -0.0217$, $p < 0.001$ between-user; $\beta = -0.0109$, $p < 0.001$ within-user), less sensitive content ($\beta = -0.0116$, $p < 0.001$ between-user; $\beta = -0.0070$, $p < 0.001$ within-user), more positive sentiment ($\beta = 0.0711$, $p < 0.001$ between-user; $\beta = 0.0370$, $p < 0.001$ within-user), and less negative sentiment ($\beta = -0.0732$, $p < 0.001$ between-user; $\beta = -0.0383$, $p < 0.001$ within-user).

Network size effects on content varied. Within-user models showed consistent suppression effects: as individuals' networks expanded, they posted less offensive ($\beta = -0.0096$, $p < 0.01$), sensitive ($\beta = -0.0110$, $p < 0.05$), and negative content ($\beta = -0.0104$, $p < 0.001$), while increasing positive expression ($\beta = 0.0208$, $p < 0.001$). Between-user models revealed more complex patterns, with larger networks associated with increased negative sentiment ($\beta = 0.0594$, $p < 0.001$) but reduced sensitive content ($\beta = -0.0547$, $p < 0.001$). Network size showed no significant main effects on offensive content between users ($\beta = -0.0004$, n.s.).

Network density showed unexpected patterns contrary to network constraint theory. Dense networks were associated with more offensive content ($\beta = 0.0697$, $p < 0.001$ between-user; $\beta = 0.0044$, $p < 0.01$ within-user) and increased negative sentiment ($\beta = 0.1183$, $p < 0.001$ between-user; $\beta = 0.0078$, $p < 0.001$ within-user), while reducing positive sentiment ($\beta = -0.0644$, $p < 0.001$ between-user). Network density showed no significant effects on sensitive content in either model.

Longitudinal trends supported behavioral adaptation hypotheses. Between-user analysis revealed declining offensive content ($\beta = -0.0120$, $p < 0.001$), sensitive content ($\beta = -0.0409$, $p < 0.001$), and negative sentiment ($\beta = -0.4769$, $p < 0.001$) over time, alongside increasing positive sentiment ($\beta = 0.0455$, $p < 0.001$). Within-user patterns showed similar moderation trends but with more complex emotional trajectories, including declining positive sentiment ($\beta = -0.0091$, $p < 0.05$) alongside declining offensive content ($\beta = -0.0143$, $p < 0.001$), sensitive content ($\beta = -0.0078$, $p < 0.05$), and negative sentiment ($\beta = -0.0133$, $p < 0.001$).

Table 2: Network Structure Effects on Tweet Content: Between-User

	Offensive	Sensitive	Positive	Negative	Neutral
Network Size	-0.0004	-0.0547***	0.0135	0.0594***	-0.0058
Network Density	0.0697***	0.0165	-0.0644***	0.1183***	-0.0186
Directedness	-0.0217***	-0.0116***	0.0711***	-0.0732***	0.0021
Time	-0.0120***	-0.0409***	0.0455***	-0.4769***	0.0825**
Constant	0.1060***	0.1392***	0.3965***	0.1637***	0.4398***
R^2	0.045	0.039	0.083	0.170	0.018
F-statistic	342.1	298.2	404.6	602.3	73.2

Significance: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

5 Discussion and Limitations

This study reveals important nuances in how social structure shapes online expression. The consistent differences between directed and

Table 3: Network Structure Effects on Tweet Content: Within-User

	Offensive	Sensitive	Positive	Negative	Neutral
Network Size	-0.0096**	-0.0110*	0.0208***	-0.0104***	-0.0103***
Network Density	0.0044**	0.0026	-0.0051	0.0078***	-0.0027
Directedness	-0.0109***	-0.0070***	0.0370***	-0.0383***	0.0014
Time	-0.0143***	-0.0078*	-0.0091*	-0.0133***	0.0224***
R^2	0.009	0.006	0.023	0.035	0.001
F-statistic	203.3	170.5	675.1	708.2	16.1

Significance: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

undirected tweets strongly support audience design theory [3, 6], demonstrating that users adjust their communication based on imagined audience characteristics even in public contexts [11]. The finding that interpersonal cues activate more restrained expression patterns suggests that perceived audience specificity, rather than actual visibility, drives behavioral adaptation.

Our findings reveal complex relationships between network properties and user expression that challenge simple constraint-based theories [4]. The contrasting effects of network size across analytical approaches suggest that while users with inherently larger networks may exhibit different baseline expression patterns, the process of network expansion within individuals consistently promotes moderation—supporting impression management theories [8]. The longitudinal decline in offensive, sensitive, and negative content provides evidence for learning-based adaptation to platform norms and audience awareness [2]. The within-user pattern of declining emotional expression overall suggests users develop more sophisticated audience management strategies over time.

The finding that dense networks facilitate rather than constrain offensive expression contradicts network constraint theory but aligns with social identity and echo chamber perspectives [4]. Dense networks may create in-group dynamics that normalize controversial expression, or the interconnectedness of contacts may reduce perceived audience diversity, leading users to feel more comfortable expressing extreme views [12] [16]. The divergent effects of network size on negative sentiment across analytical approaches highlight the importance of distinguishing selection from influence processes. Users who naturally express more negative sentiment may attract larger networks, while network expansion within individuals promotes emotional moderation—suggesting both homophily and adaptation mechanisms operate simultaneously.

While these patterns are consistent with theoretical accounts of norm activation, accountability, and audience perception, it does not enable causal interpretation. All relationships observed are correlational. Moreover, the classification of tweet content—based on NLP models—may miss contextual nuance. Our network measures are based on reciprocated @mentions, which reflect social interaction but may not capture deeper relational ties or visibility across the feed. The data filtering to users with stable, active accounts may bias toward more strategic or audience-aware users, potentially limiting generalizability to users who engage in sporadic social media use.

Despite these limitations, the consistency and interpretability of our results suggest that users navigate public online expression by structuring *how* and *to whom* they speak. Message directedness

and broader social context jointly shape expressive behavior—even when formal visibility remains constant.

6 Implications and Conclusion

Our findings suggest several design strategies for supporting more thoughtful communication in online environments. The consistent moderating effects of message directedness suggest that platforms could promote more civil discourse by emphasizing conversational context. Features such as enhanced reply threading, conversation previews, or visual indicators of social connections could help users maintain awareness of specific audience members rather than treating all posts as broadcast communication to an ambiguous audience. Understanding how network properties influence expression could inform content moderation strategies. Platforms might consider network context when assessing potentially harmful content—for instance, flagging undirected posts from users embedded in dense networks as potentially higher-risk for norm violations. Design features that help users visualize audience composition or relationship strength could support more intentional self-presentation. Rather than relying solely on privacy controls, platforms could provide tools for understanding and managing audience complexity.

Our results suggest that content may carry different risks depending on network position and message directedness. Moderation algorithms could incorporate these structural factors to improve accuracy and reduce false positives. Understanding that certain network configurations (large, sparse networks) promote restraint while others (dense networks) may facilitate extreme expression could inform proactive intervention strategies.

Rather than viewing context collapse as a uniform constraint, our findings suggest it operates differently across network positions and interaction patterns. Users may experience varying degrees of context collapse based on their structural position and communication strategies. The study demonstrates that online behavior is shaped by multiple levels of social context simultaneously—from specific interaction partners to broader network structure.

Future work should investigate the causal mechanisms underlying network effects on expression, potentially through natural experiments or intervention studies that manipulate network structure or audience salience. Testing these relationships across different platforms with varying affordances and user bases would strengthen generalizability and reveal platform-specific effects. Combining behavioral data with qualitative investigations of user perceptions and strategies could provide deeper insight into the psychological mechanisms driving observed patterns.

This study demonstrates that online expression is systematically shaped by both interactional context and network structure, even in seemingly undifferentiated public spaces. Users navigate context collapse by strategically adjusting their communication style based on audience structure and message targeting. These findings highlight the continued relevance of social norms and audience awareness in online environments. By recognizing how structural and interactional cues influence expression, online platforms can better support users in managing audience complexity while promoting more thoughtful and prosocial communication norms.

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